Name :- Kartik Ram Birajdar

Roll No:- A-13

Branch:- AI&DS

Assignment :-10

Aim – Version Control using Git & GitHub

Theory:-

Git is a distributed version control system used to track changes in source code during software development.

GitHub is a cloud-based platform that hosts Git repositories and enables collaboration among developers.

Version control helps:- Track changes Collaborate with others Restore previous versions Manage project history

Step-by-Step Procedure

1. Install Git Download Git from official website Install with default settings **2.**

Configure Username & Email git config --global user.name "Vaibhav Gadhave" git

config --global user.email "vaibhavgadhave646@gmail.com" **3. Create Local Repository**

mkdir myproject cd myproject git init

4. **Create Files**echo "Hello Git" > file.txt

5. **Add Files to Staging Area**

git add file.txt

6. **Commit Changes**

git commit -m "Initial commit"

7. **Create GitHub Repository**Go to GitHub website

- Click New Repository
- • Enter name and create report

8. Link Local Repo to GitHub

git remote add origin https://github.com/username/repo-name.git

Screenshots to Include

You should attach screenshots of:Git

Installation Verification git --version

Repository Creation

git init

Commit History git

log

Commit History

git log

GitHub Repository Page Show created repo in browser Successful Push Terminal showing push success Commit Logs git log --oneline

Files on GitHub

Show uploaded file in repo **Questions &**

Answers

Git	GitHub
Version control system	Hosting platform
Works locally	Works online
Tracks changes	Stores repositories
CLI-based tool	GUI + web interface

2. Difference between git add and git commit

git add	git commit
Moves files to staging area	Saves changes permanently
Prepares files for commit	Records snapshot
Temporary step	Final step

3. What is Staging Area in Git?

The staging area is an intermediate area where changes are prepared before committing. It allows:

- Selecting specific changes Reviewing
- before commit

4. What happens if you modify a file after committing it?

- Git detects changes File
- becomes modified You
- must: git add again git
- commit again
-
- **5. Difference between git push and git pull**

git push	git pull
Uploads local changes to GitHub	Downloads changes from GitHub

Local → Remote	Remote → Local	Used after commit	Used to update project

6. What is Version Control?

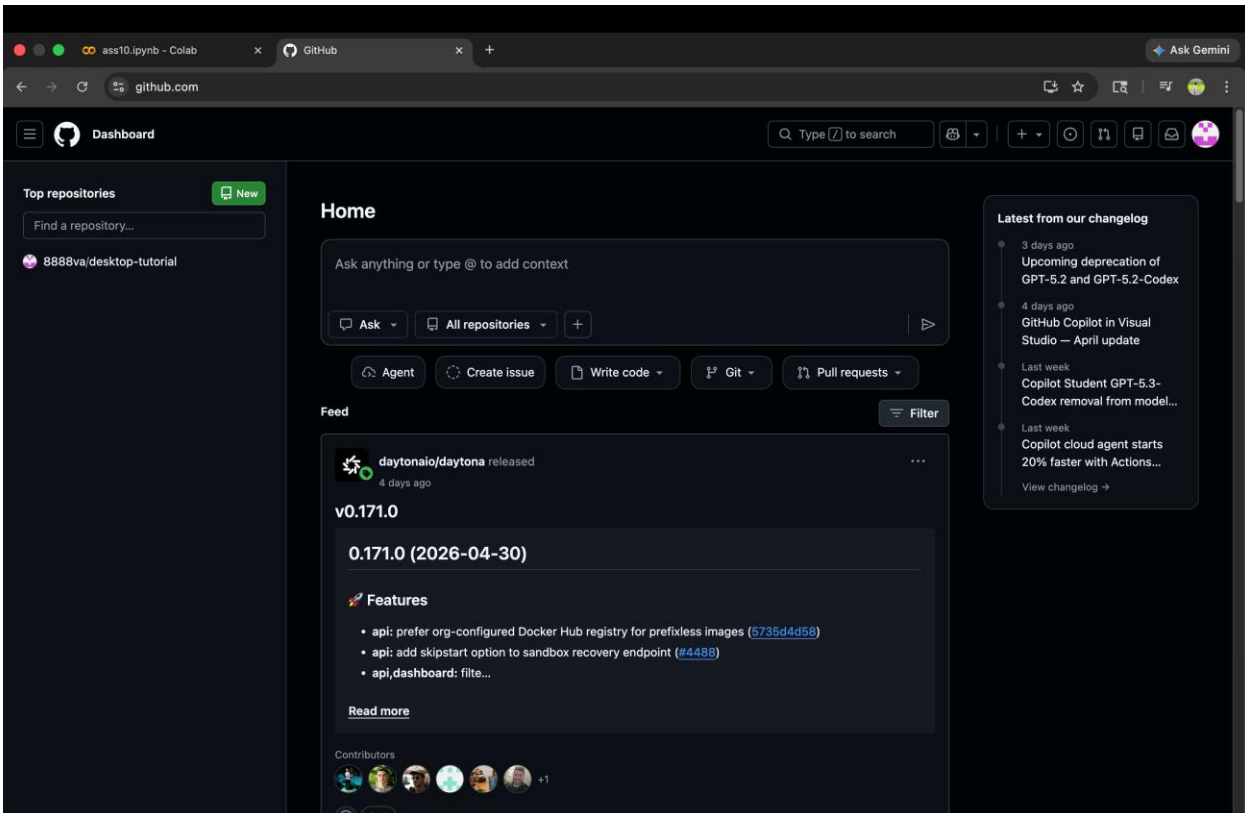
Centralized VCS	Distributed VCS
Single central server	Multiple copies (local repos)
Requires internet	Works offline
Example: SVN	Example: Git
Less flexible	Highly flexible

```
git push origin main
Push successful
```

```
git log
commit abc123
Author: User
Message: Initial commit
```

git init
Initialized empty Git repository

git --version
Output: git version 2.43.0


ass10.ipynb - Colab
GitHub
Ask Gemini
github.com
Dashboard
Type / to search
Top repositories
New
Find a repository...
8888va/desktop-tutorial
Home
Ask anything or type @ to add context
Ask
All repositories
Agent
Create issue
Write code
Git
Pull requests
Feed
Filter
daytonaio/daytona released
4 days ago
v0.171.0
0.171.0 (2026-04-30)
Features
api: prefer org-configured Docker Hub registry for prefixless images (5735d4d58)
api: add skipstart option to sandbox recovery endpoint (#4488)
api,dashboard: filte...
Read more
Contributors
Latest from our changelog
3 days ago
Upcoming deprecation of GPT-5.2 and GPT-5.2-Codex
4 days ago
GitHub Copilot in Visual Studio — April update
Last week
Copilot Student GPT-5.3-Codex removal from model...
Last week
Copilot cloud agent starts 20% faster with Actions...
View changelog →